

SECTION A

QUESTION 1 (COMPULSORY)

XYZ Corporation, a medium-sized company, recently experienced a significant increase in network traffic due to the expansion of its online services. As the lead server administrator, you are tasked with addressing the challenges associated with this surge in traffic and ensuring optimal performance and availability of the company's servers.

- A. Outline the steps you would take to assess the impact of the increased network traffic on the company's servers. Consider factors such as server resource utilization, network bandwidth, and application performance. Propose specific monitoring tools and metrics you would use to gather relevant data for your assessment.

[5 Marks]

- B. After analysing the impact of the increased network traffic, identify potential bottlenecks and performance issues that may be affecting the company's servers. Describe the techniques and strategies you would employ to troubleshoot and diagnose these issues effectively. Include considerations for both hardware and software components of the server infrastructure.

[5 Marks]

SECTION B

QUESTION 2

Institutions deploy availability mechanisms to ensure authorized users can access information as needed for their success. It's crucial to uphold operational computing systems, secure controls, and communication channels. As a prospective System Administrator, recommend five (5) solutions for the institution to implement in order to ensure availability.

[10 Marks]

QUESTION 3

Imagine you're the technical advisor for the University of Kigali. Offer guidance on the concept, challenges and advantages of implementing the following techniques.

- A. Clustering technique

[5 Marks]

- B. Virtualization technology

[5 Marks]

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